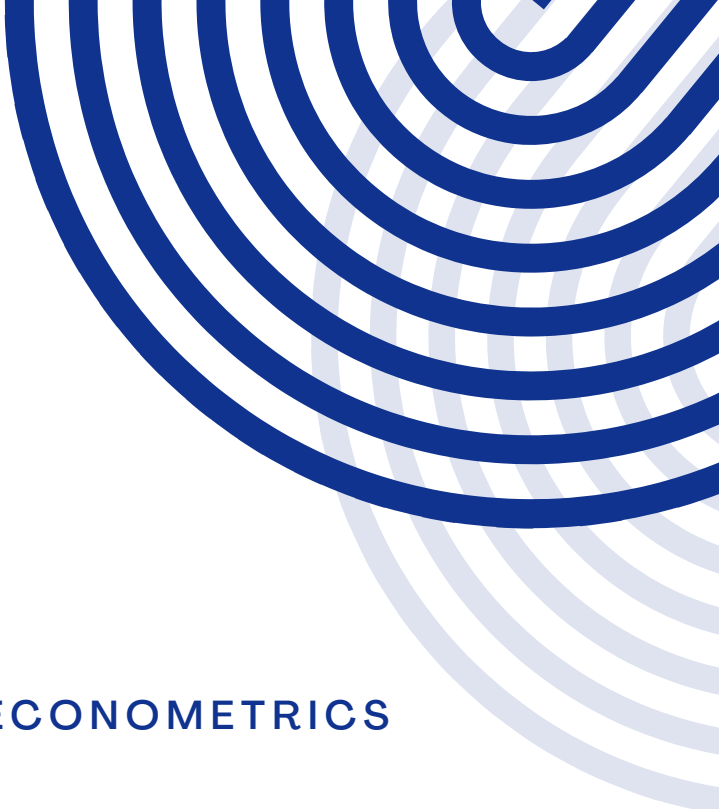




ECON712: APPLIED ECONOMETRICS

Research Project:
**“Inflation and Stock Returns: A Firm-Level
Analysis of Macroeconomic Dynamics
Across Key Sectors in the S&P 500”**

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Introduction

The relationship between stock returns and inflation has been a subject of academic inquiry for decades, with research interest fluctuating in response to the macroeconomic significance of inflation. The high inflation episodes of the 1970s severely impacted living standards and economic stability, leading to poor stock market performance and prompting extensive studies into the connection between stock returns and inflation. More recently, the COVID-19 pandemic has reignited interest in this topic. In response to the pandemic, central banks and finance ministries implemented unprecedented policy measures to prevent economic collapse and support recovery. While these actions were critical in averting a global economic downturn, they have also raised concerns about potential side effects, such as asset bubbles and growing inequality. As Consumer Price Indices rise sharply, questions persist about whether this inflationary trend is temporary or permanent, and its implications for financial markets.

The study of stock returns and inflation lies at the intersection of finance and macroeconomics. Financial economists examine this relationship to assess the extent to which stocks can hedge against inflation risk, a critical consideration for trading strategies and risk management. On the other hand, monetary economists explore the potential influence of inflation and monetary policy on stock returns. Since the stock market serves as a vital channel for corporate financing, fluctuations in stock prices have significant implications for real economic activity and financial stability. These include effects on corporate borrowing, investment decisions, and household consumption and saving behavior, driven by changes in net worth (Zhang, Z. (2021). Consequently, central banks with a mandate to stabilize prices must consider the impact of inflation on asset returns when formulating policy.

Historically, rising inflation in developed countries, especially in US, has been associated with a decline in real stock prices. Equity shares, which represent claims on the future output of firms, have not proven to be an effective hedge against inflation risk. This intriguing phenomenon has prompted economists to explore why inflation, a nominal variable, influences real stock prices and the real value of claims on physical assets.

Several theories have been proposed to explain the negative correlation between stock returns and inflation, with monetary economics offering one prominent explanation. According to these hypotheses, the negative relationship stems from central banks' countercyclical policy responses. When inflation rises, central banks aiming to maintain price stability often increase their policy rates. This adjustment affects real interest rates, which, in turn, influence stock prices.

Since stock prices reflect the present value of all future cash flows, a rise in interest rates (used as the discount rate) reduces the net present value of stocks. Additionally, higher interest rates raise borrowing costs for firms, enhance the appeal of alternative investments like bonds and deposits, reduce liquidity in the stock market, and exert downward pressure on stock returns.

Previous research on the relationship between inflation and stock performance has predominantly concentrated on broad analyses of single countries (e.g., Fama, 1981) or small groups of industrialized nations, often examining aggregate market indices rather than individual firms. My research seeks to address this gap by focusing on the nuanced effects of inflation on individual firms within the S&P 500 index combined with thorough analysis of US macroeconomic variables. Specifically, I will explore how inflation impacts stock performance across key sectors—energy, IT, and health care—each of which exhibits unique characteristics and sensitivities to inflationary pressures. Additionally, I will analyze the effects of industry-specific inflation rates on the stock prices of corresponding industries, recognizing that sector-specific inflation dynamics, such as rising energy costs or fluctuating healthcare expenditures, can have distinct implications for profitability and stock valuations. Given the economic significance of inflation and its potential to influence corporate performance through factors like input costs, pricing power, and demand elasticity, a sector-specific, firm-level, and industry-inflation perspective is essential. This approach will provide valuable insights for investors managing inflation risk in diversified portfolios and for policymakers seeking to understand the broader economic implications of sectoral responses to both general and industry-specific inflation. Moreover, this study emphasizes the monthly price sensitivity of stock prices, ensuring that all variables, including inflation, macroeconomic controls, and stock returns, are analyzed on a monthly basis. By adopting a consistent temporal frequency, the research captures the short-term dynamics between inflation and stock performance, providing insights into how monthly fluctuations in inflation—both general and industry-specific—affect stock valuations in the short term. This focus on monthly data is particularly relevant for identifying patterns and relationships that may be obscured in longer time horizons, such as annual analyses. It also aligns with practical considerations for investors and policymakers who often make decisions based on monthly updates of economic and financial indicators. This granularity enhances the relevance of the findings for stakeholders looking to navigate inflationary pressures with precision and timeliness. By combining these layers of analysis, my study offers a unique and actionable contribution to the literature on inflation and financial markets.

Building on the foundational frameworks established in prior research, my analysis incorporates classic control variables to investigate the relationship between inflation and stock performance.

The first control variable is the expected inflation rate which might be actually identifies as the primary independent variable, since financial analyst usually adjust their stock prices depending on the expected inflation rate the following year. Drawing from seminal works by Fama (1981), Mondelker (1985), Kaul (1987), and Zhang (2021), I integrate key economic indicators into my model. These include measures of real activity (gross national product and industrial production), interest rates, three-month Treasury bill rates, and money supply growth (M2 growth rate). Moreover, I will take into account for fixed effect of industries. The inclusion of these variables reflects the influence of monetary and real economic factors on stock performance.

By controlling for these well-established determinants, my research ensures a robust framework for isolating the specific impacts of both general and industry-specific inflation on stock prices. This approach not only aligns with existing methodologies but also provides a comprehensive basis for examining how inflation interacts with broader macroeconomic forces to shape sectoral and firm-level stock returns.

Literature Review

The relationship between stock returns, inflation, and macroeconomic variables has been a focal point of financial economics, explored extensively in empirical and theoretical frameworks. Fama (1981) proposed the *proxy hypothesis*, asserting that the negative correlation between stock returns and inflation is driven by the interaction between inflation and real economic activity. Fama's findings suggest that both stock returns and inflation are indirectly linked through real activity, with inflation negatively correlated with real activity and stock returns positively correlated. By controlling for real variables such as GNP and industrial production, the anomaly of a negative stock return-inflation relationship diminishes, aligning stock returns with fundamental economic indicators (Stock returns, real activity, inflation, and money).

Building on Fama's framework, Mandelker and Tandon (1985) examined this relationship across six major industrialized countries. Their findings corroborate the proxy hypothesis, demonstrating that real activity variables dominate in explaining stock returns, while money growth rates and inflation play a secondary role. The authors emphasize that inflation's adverse effects on stock returns are contingent upon its negative influence on real activity, reinforcing the relevance of integrating macroeconomic variables like interest rates and monetary growth in such analyses (Common stock returns, real activity, money, and inflation: Some international evidence).

Kaul (1987) further elaborated on the monetary sector's role, positing that the negative stock return-inflation relationship is influenced by the equilibrium processes in money demand and supply. His work highlights that countercyclical monetary policy exacerbates this negative relationship by affecting inflation expectations and real economic activity. However, Kaul also noted that procyclical monetary policy could reverse this effect, demonstrating the temporal variability in stock return-inflation dynamics (Stock returns and inflation: The role of the monetary sector).

In more recent work, Zhang (2021) expanded the scope of analysis to 71 advanced and emerging economies, illustrating how variations in monetary policy regimes shape the stock return-inflation relationship. Zhang demonstrated that in inflation-targeting economies, stock markets are highly sensitive to inflation, particularly when inflation exceeds target bands. Conversely, in economies with less flexible monetary policies or exchange rate anchors, this sensitivity is diminished. This study underscores the critical role of monetary policy frameworks in moderating the inflation-stock return nexus (Inflation and stock returns in inflation-targeting economies).

Eldomiaty, T. I., Soliman, A. A., Fikri, A. M., & Anis, M. S. (2019) contributed to the discourse by analyzing inflation's effect on stock prices within the context of real interest rates. Using the stock duration model, their findings confirm a persistent negative relationship between inflation and stock prices while highlighting that real interest rates have a positive influence. Their work differentiates between the effects of inflation and real interest rates, providing nuanced insights into how these variables interact with equity valuations (Inflation and stock returns: Empirical evidence from sectoral data).

Collectively, these studies underscore the complex interplay between inflation, real activity, and monetary policy in determining stock returns. While the negative stock return-inflation relationship is well-documented, its underlying mechanisms vary across time, regions, and monetary regimes. This body of literature informs my research by providing a robust framework for examining how inflation impacts firm-level stock performance across different sectors within the S&P 500, with particular attention to the moderating role of industry-specific inflation rates.

Data summary.

The dataset for this study comprises monthly stock prices and returns for the S&P 500 index, covering a period of 10 years, up to the present or to be more specific from Dec 2014 till June 2024. The reason why I didn't encompass further month is because some macroeconomic indicators which I utilize as control variables are not available yet. Also, this timeframe was chosen primarily because historical stock prices on Nasdaq—the most reliable data source for U.S. stock prices—are available for a maximum of 10 years. Additionally, Nasdaq offers free access to downloadable historical data, which provides an accessible and reliable dataset.

Fama (1981) emphasizes that the relationship between stock returns and inflation is fundamentally tied to real economic activity. Adjusting nominal returns for inflation ensures that the analysis isolates the real factors driving stock performance, rather than conflating them with inflationary noise. Similarly, Mandelker and Tandon (1985) highlight the importance of using real returns to uncover the underlying dynamics between stock performance, inflation, and macroeconomic variables, noting that nominal returns can obscure these relationships.

Monthly inflation rates were sourced from the U.S. Bureau of Labor Statistics (BLS), recognized as the most reliable authority on inflation data in the United States. By using BLS data, I ensure that the inflation metrics in this analysis are consistent, high-quality, and accurate.

Furthermore, I include the expected monthly inflation rate as a control variable, which is extracted Federal Reserve Bank of St. Louis (FRED), drawing on the foundational works of Fama (1981) and Mandelker and Tandon (1985). Expected inflation is crucial because it captures market anticipations regarding future price levels, which influence investment decisions and valuation processes. Fama (1981) argues that inflation expectations, as a component of monetary conditions, are intertwined with real economic activity, a key driver of stock performance. Similarly, Mandelker and Tandon (1985) demonstrate that expected inflation can serve as a predictive indicator for real stock returns, highlighting its significance in disentangling the complex relationship between stock performance, inflation, and real economic variables. By incorporating expected inflation, this study accounts for its potential to affect stock returns through both direct channels, such as discount rates, and indirect channels, such as changes in real activity influenced by monetary policy responses.

A unique aspect of this research is the inclusion of industry-specific inflation rates to analyze their impact on corresponding industry stocks. For the Energy and Health Care industries, inflation rates were sourced from the U.S. Bureau of Labor Statistics (BLS), which provides

detailed breakdowns of inflation by industry. For the IT industry, the data was derived from the Federal Reserve Economic Data (FRED), specifically the "Consumer Price Index for All Urban Consumers: Information Technology, Hardware, and Services in U.S. City Average (CUSR0000SEEE)." These industry-specific inflation rates were incorporated into the analysis by repeating them only for companies within their respective industries, while a fixed-effects approach accounted for additional unobserved industry-level heterogeneity. This method ensures that the distinct inflationary pressures faced by each sector are accurately represented, recognizing that industries like Energy, IT, and Health Care experience unique cost structures, input dependencies, and demand sensitivities. By aligning industry-specific inflation rates with the stocks in those industries, the study provides a granular perspective on how inflation dynamics influence stock performance, thereby offering a novel contribution to the literature.

Additionally, this study incorporates M2 money supply as a control variable (Source FRED), as advocated by Fama (1981) and Mandelker and Tandon (1985). M2, a broad measure of money supply that includes cash, checking deposits, and easily convertible near money, is a critical variable for understanding the monetary environment's influence on stock performance. According to Fama (1981), variations in money supply influence inflationary expectations and real economic activity, both of which significantly affect equity valuations. Mandelker and Tandon (1985) similarly highlight that money supply growth often interacts with inflation and real activity, making it a vital factor in explaining stock returns. Including M2 ensures that monetary influences on stock performance are adequately captured in the analysis.

The study also incorporates the three-month Treasury bill rate (Source FRED) as a control variable instead of a broader interest rate measure. This choice is justified based on the works of Mandelker and Tandon (1985), who demonstrate that short-term interest rates, like the three-month Treasury bill, are reliable predictors of expected inflation and reflect the opportunity cost of capital for short-term investments. Fama (1981) further supports this selection by highlighting the Treasury bill rate's role in capturing the risk-free rate used in discounting future cash flows, directly influencing stock valuations. This specificity makes the three-month Treasury bill rate particularly relevant for examining the interplay between monetary policy, inflation, and stock performance.

Real activity, measured by Gross National Product (GNP) and industrial production (Source FRED), is another essential control variable in this study. Fama (1981) posits that stock returns are positively correlated with real economic variables, as they reflect the underlying profitability and growth potential of firms. Mandelker and Tandon (1985) confirm this relationship,

emphasizing that real activity variables dominate monetary factors in explaining stock returns. Including GNP and industrial production allows this study to capture the effects of real economic growth on equity performance, ensuring a comprehensive analysis of the factors driving stock returns. It should be noted GNP is the only variables I have in my research which is published quarterly. Hence, I have applied quarterly data for all months within the quarter they belong for the sake of the consistency of the data.

What distinguishes this research from similar studies is its focus on analyzing the impact of macroeconomic variables on specific industries within the S&P 500. More specifically, it examines three primary industries within the index: energy, IT, and health care. By incorporating industry-specific fixed effects through dummy variables for IT and health care, this study aims to isolate the differential impacts of macroeconomic factors across industries.

Regarding the industries analyzed, the following companies represent the focus of this study within these sectors:

Energy Companies	IT Companies	Health Care Companies
Exxon Mobil Corporation (XOM)	Apple Inc. (AAPL)	Eli Lilly and Company (LLY)
Chevron Corporation (CVX)	NVIDIA Corporation (NVDA)	UnitedHealth Group Incorporated (UNH)
ConocoPhillips (COP)	Microsoft Corporation (MSFT)	Johnson & Johnson (JNJ)
EOG Resources, Inc. (EOG)	Broadcom Inc. (AVGO)	AbbVie Inc. (ABBV)
The Williams Companies, Inc. (WMB)	Oracle Corporation (ORCL)	Merck & Co., Inc. (MRK)
Schlumberger Limited (SLB)	Adobe Inc. (ADBE)	Thermo Fisher Scientific Inc. (TMO)
Kinder Morgan, Inc. (KMI)	Salesforce, Inc. (CRM)	Abbott Laboratories (ABT)
Diamondback Energy, Inc. (FANG)	Cisco Systems, Inc. (CSCO)	Pfizer Inc. (PFE)
Phillips 66 (PSX)	Accenture plc (CAN)	Amgen Inc. (AMGN)
Marathon Petroleum Corporation (MPC)	Texas Instruments Incorporated (TXN)	Bristol-Myers Squibb Company (BMY)
Occidental Petroleum Corporation (OXY)		
Hess Corporation (HES)		
Valero Energy Corporation (VLO)		
Targa Resources Corp. (TRGP)		
Halliburton Company (HAL)		
Devon Energy Corporation (DVN)		
EQT Corporation (EQT)		
Coterra Energy Inc. (CTRA)		

Marathon Oil Corporation (MRO)		
APA Corporation (APA)		
Overall, 40 companies with each having 115 observations (months), in total 4600 observations		

The companies were chosen based on their market capitalization to ensure that the sample is representative of each sector's influence on the S&P 500. The larger the market cap, the greater the influence on the index, thus providing a balanced view of each industry's reaction to inflation.

Here are some basic facts about the independent variables:

Variable	Obs	Mean	Std. Dev.
Monthly Real Inflation (I)	115	2.874783	2.347221
Monthly Expected Inflation (EI)	115	2.081326	0.2074269
3-month T-bill return (TB)	115	1.653757	1.855046
Money Supply Growth (M2)	115	0.5206279	0.8792511
Monthly Industries Production change in % (DPR)	115	0.0111512	1.630278
Gross Nation Product change in % (GNP)	115	0.4208529	1.383908
Monthly Inflation rate in IT industry (I_IT)	115	7.467904	0.304626
Monthly Inflation rate in energy industry (I_enrg)	115	2.486957	14.37449
Monthly Inflation rate in Health Care industry (I_HC)	115	2.767826	1.745347

Empirical Strategy

To analyze the impact of macroeconomic variables on stock returns across industries, this study employs a panel data regression framework. The dataset is structured in a long format, featuring monthly observations for each company from December 2014 to June 2024. The dependent variable is real stock return, derived by converting nominal returns to real returns through inflation adjustment. This conversion is critical for capturing the true economic value of stock performance, as nominal returns fail to account for the eroding effects of inflation on purchasing power. By focusing on real returns, the analysis offers a more accurate reflection of wealth changes for investors.

The independent variables include real inflation, expected inflation, industry-specific inflation, T-bill rates, money supply, industrial production, and Gross National Product (GNP). All

variables are expressed in percentage change form to ensure consistency and comparability, given the differing scales of stock prices and macroeconomic indicators. This transformation standardizes the data, facilitating direct comparison of the magnitude and direction of changes and enabling more interpretable relationships between variables.

Since the stock price data is originally provided on a daily basis, monthly stock prices are derived by selecting the closing price on the last trading day of each month. This method simplifies the dataset while ensuring consistency and accuracy for monthly analysis.

To capture industry-specific effects, dummy variables are introduced for the IT and health care industries, with the energy industry serving as the baseline. Additionally, industry-specific inflation rates are included to account for inflationary dynamics unique to each sector. The analysis employs a fixed-effects model to control for unobserved heterogeneity across companies, ensuring that time-invariant factors specific to each company are accounted for. Robust standard errors clustered at the company level are applied to address potential heteroskedasticity and within-panel correlation, improving the reliability of the coefficient estimates.

This methodology allows for an in-depth exploration of how macroeconomic variables, including inflation, influence stock returns across three primary industries: IT, health care, and energy. By combining fixed effects, robust standard errors, and industry-specific inflation rates, the model evaluates whether the relationship between stock returns and macroeconomic variables varies by sector.

Finally, the analysis assumes that macroeconomic variables such as inflation and expected inflation are exogenous or predetermined with respect to real stock returns. This assumption aligns with prior research (e.g., Fama, 1981; Kaul, 1987), which treats macroeconomic factors as external drivers of asset prices. However, the potential for endogeneity remains, particularly for variables like expected inflation, which may be influenced by contemporaneous market dynamics. Addressing this limitation, future research could employ instrumental variable techniques or dynamic modeling approaches to better account for simultaneity and enhance the robustness of causal inferences.

Main Results

The OLS is run using the following code on STATA: `xtreg real_return I EI I_IS TB M2 DPR GNP IT HC, robust`

This code runs a panel data regression (`xtreg`) to analyze how macroeconomic variables like inflation (I, EI), Treasury bill rates (TB), and industry-specific factors impact real stock returns (`real_return`). By including dummy variables for IT and Health Care (IT, HC), it accounts for sectoral differences, with Energy as the baseline. The use of `robust` ensures the standard errors are adjusted for heteroskedasticity and within-panel correlation, providing more reliable and accurate inference for the model's coefficients.

```
. xtreg real_return I EI I_IS TB M2 DPR GNP IT HC , robust

Random-effects GLS regression              Number of obs      =       4600
Group variable: company_id                Number of groups   =        40

R-sq:  within = 0.1097                    Obs per group: min =       115
       between = 0.2439                      avg      =      115.0
       overall = 0.1104                      max      =       115

                                           Wald chi2(9)       =      124.06
corr(u_i, X)   = 0 (assumed)              Prob > chi2        =       0.0000
```

(Std. Err. adjusted for 40 clusters in company_id)

real_return	Coef.	Robust Std. Err.	z	P> z	[95% Conf. Interval]	
I	-.118444	.0550364	-2.15	0.031	-.2263134	-.0105745
EI	1.246297	.9003613	1.38	0.166	-.5183788	3.010973
I_IS	.0028812	.012272	0.23	0.814	-.0211715	.0269338
TB	.1487219	.0502203	2.96	0.003	.050292	.2471518
M2	1.79233	.4474597	4.01	0.000	.9153247	2.669335
DPR	1.407929	.2535073	5.55	0.000	.911064	1.904794
GNP	.0044729	.0678256	0.07	0.947	-.1284628	.1374085
IT	.4955366	.1841152	2.69	0.007	.1346774	.8563957
HC	.0971847	.1067498	0.91	0.363	-.112041	.3064105
_cons	-3.852376	2.133019	-1.81	0.071	-8.033017	.3282651
sigma_u	0					
sigma_e	5.7851903					
rho	0	(fraction of variance due to u_i)				

Let's analyze this result of the regression:

The R-squared (within) value of 0.1097 measures the proportion of variation in the dependent variable, real stock returns, that is explained by the independent variables (I, EI, I_IS, TB, M2, DPR, GNP) after accounting for company-specific effects. In this case, approximately 10.97% of the variation in real stock returns within individual companies over time is explained by the model. While this value might appear low, it is typical for financial datasets, particularly in panel data analysis, where much of the variation in stock returns is influenced by unpredictable market dynamics, idiosyncratic shocks, or omitted variables.

In the context of financial research, R-squared values in the range of 5-15% are common and still provide meaningful insights. The "within" R-squared specifically evaluates how well the model explains variations that occur within companies over time, making it well-suited for panel regression approaches like random effects.

The Wald Chi-square test evaluates whether the independent variables in the model collectively have a statistically significant relationship with the dependent variable, real stock returns. In this case, the Wald Chi-square value of 124.06, coupled with a p-value of 0.0000, indicates that the model is highly statistically significant at any conventional significance level (e.g., 1%, 5%, or 10%). This result means that the inclusion of macroeconomic variables like inflation, Treasury Bill rates, money supply, and industrial production significantly improves the model's ability to explain real stock returns compared to a model with no predictors. The Wald test reinforces the conclusion that at least one of the independent variables has a meaningful and statistically significant effect on real stock returns.

Real Inflation (I): The coefficient for real inflation is -0.118444 and statistically significant (p-value = 0.031), indicating a negative relationship between inflation and stock returns. This finding supports the theoretical framework proposed by Fama (1981), which suggests that inflation adversely impacts stock returns by reducing real economic activity and purchasing power. The negative relationship highlights how inflationary pressures erode the real value of future cash flows, thereby reducing stock valuations. This is consistent with the “proxy hypothesis,” which links inflation to diminished real economic performance. However, the magnitude of this effect suggests that the relationship may vary depending on industry and market conditions, making inflation a critical variable to monitor in portfolio risk management.

Expected Inflation (EI): The coefficient for expected inflation is 1.2463, but it is not statistically significant (p-value = 0.166). While the positive coefficient aligns with the Fisher Hypothesis, which suggests that expected inflation is fully incorporated into nominal returns, the

lack of statistical significance implies that expected inflation may not play a major role in influencing real stock returns in this dataset. This result differs from prior studies emphasizing the strong influence of expected inflation on equity markets. One possible explanation is that the market participants in the observed sample period effectively anticipated and priced in expected inflation, thereby reducing its measurable impact on returns. Alternatively, the weak significance could reflect noise in the estimation of expected inflation itself.

Industry-Specific Inflation (I_IS): The coefficient for industry-specific inflation is close to zero (0.00288) and not statistically significant (p-value = 0.814), suggesting that inflation specific to individual industries does not significantly influence real stock returns. This finding could indicate that industry-level inflationary effects are either offset by other macroeconomic factors or that firms within the same industry are equally affected, rendering the impact negligible in cross-sectional analyses. The absence of a significant relationship with industry-specific inflation contrasts with some sector-specific studies that highlight the role of localized price pressures, such as Eldomiaty (2019), who found inflation impacts to vary by industry.

Treasury Bill Rates (TB): The regression reveals a positive and statistically significant relationship between Treasury Bill rates and real stock returns, with a coefficient of 0.14872 (p-value = 0.003). This indicates that higher short-term interest rates are associated with higher real stock returns. This finding diverges from classical theories that often predict a negative relationship, as higher rates increase the cost of borrowing and reduce equity valuations. In this context, the positive relationship could reflect the broader macroeconomic environment, where rising interest rates signal improved economic conditions and investor confidence, thereby boosting stock returns. This result aligns partially with studies highlighting the complex relationship between short-term rates and equity performance during periods of economic expansion.

Money Supply (M2): The coefficient for money supply is 1.7923 and highly significant (p-value = 0.000), indicating a robust positive impact on real stock returns. This result supports the theoretical linkage between liquidity and equity markets, where an increase in money supply enhances market liquidity, reduces borrowing costs, and encourages investment. The strong positive relationship aligns with the findings of Eldomiaty (2019), which emphasize the role of liquidity in driving stock market performance. However, it also contrasts with some studies that warn about speculative bubbles fueled by excess liquidity, underscoring the need for further research on money supply's dual role.

Industrial Production (DPR): Industrial production exhibits a strong and positive relationship with real stock returns, with a coefficient of 1.40793 (p-value = 0.000). This finding aligns closely with Fama's (1981) proxy hypothesis, which posits that real economic activity is a primary determinant of stock returns. Higher industrial production typically signals economic growth, improving corporate profitability and boosting investor sentiment. The significant and positive impact of industrial production underscores its importance as a key macroeconomic indicator for equity markets.

Gross National Product (GNP): The coefficient for GNP is near zero (0.00447) and not statistically significant (p-value = 0.947), suggesting that GNP does not have a measurable impact on stock returns in this model. This result may reflect the broad nature of GNP as an economic indicator, which could mask more nuanced industry-specific effects. Alternatively, other included variables, such as industrial production and money supply, may better capture the relevant economic dynamics influencing stock returns.

Industry Fixed Effects (IT and HC): The coefficient for the IT dummy is 0.4954 and statistically significant (p=0.007), indicating that IT sector companies, on average, exhibit 49.54 basis points higher real stock returns compared to the Energy sector, holding other variables constant. This highlights the IT sector's responsiveness to macroeconomic conditions, likely driven by its reliance on innovation cycles and sensitivity to interest rates.

In contrast, the coefficient for the Health Care dummy is 0.0972 but statistically insignificant (p=0.645), suggesting no meaningful difference in real stock returns between the Health Care and Energy sectors. This lack of significance aligns with the Health Care sector's relative stability, often insulated from economic volatility due to consistent demand and regulatory frameworks.

These findings emphasize the model's improved ability to distinguish within-industry effects while preserving between-industry differences. The inclusion of statistically significant IT dummy effects underscores the unique role of macroeconomic dynamics in shaping stock returns for growth-driven sectors like IT, while the absence of a significant effect for Health Care reflects its more stable and defensive nature. These numeric insights reinforce the nuanced understanding of sectoral behaviors under varying economic conditions.

The results provide mixed support for existing theories. The negative impact of real inflation aligns with the proxy hypothesis and the notion that inflation reduces real cash flows and stock valuations. However, the weak relationship between expected inflation and stock returns

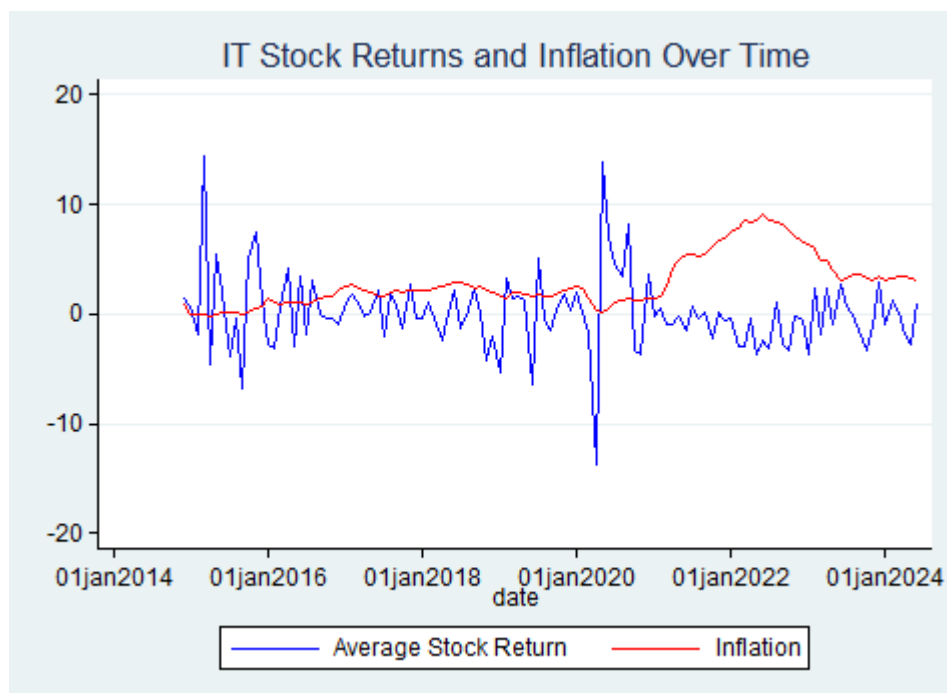
contrasts with theories, such as the Fisher Hypothesis, that predict a significant positive effect. The positive impact of Treasury Bill rates and money supply highlights the nuanced role of monetary policy in shaping equity markets. The strong influence of industrial production reaffirms the critical link between real economic activity and stock market performance.

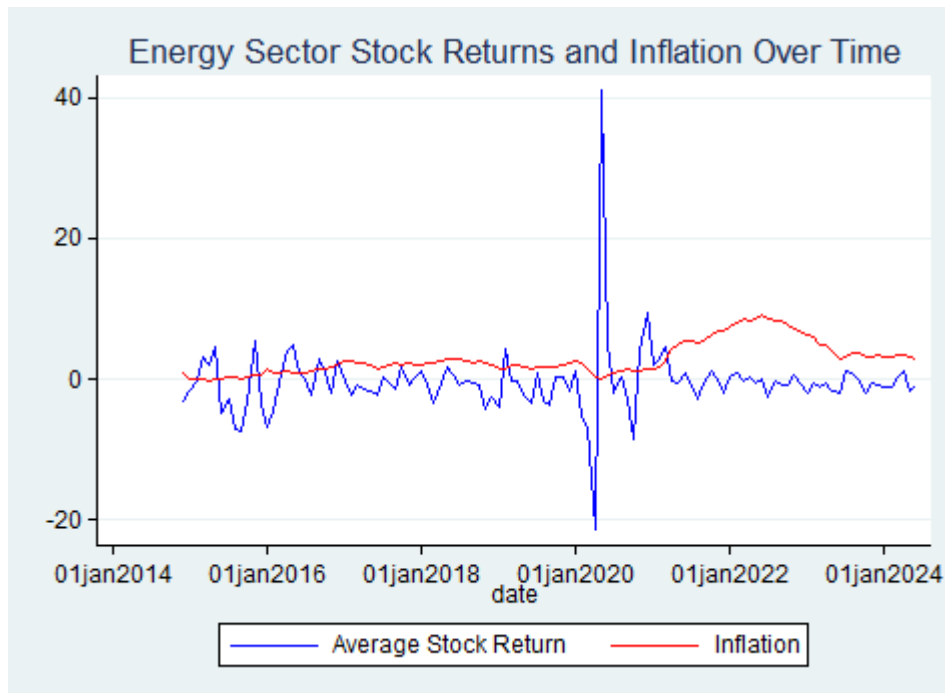
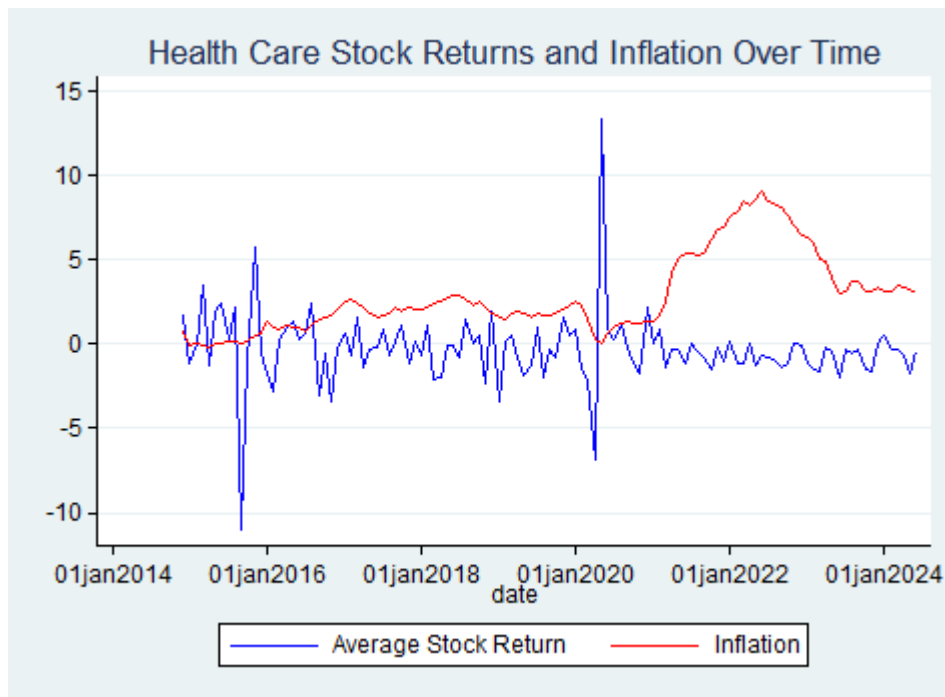
Overall, my model in accordance with the results of my regression analysis looks as follows:

$$\text{real_return} = -3.852 + (-0.118 \cdot I) + (1.246 \cdot EI) + (0.003 \cdot I_IS) + (0.148 \cdot TB) + (1.792 \cdot M2) + (1.407 \cdot DP R) + (0.004 \cdot GNP) + (0.495 \cdot IT) + (0.097 \cdot HC) + \epsilon$$

ϵ : Captures the unobserved factors affecting real stock returns that are not included in the model.

In the light of these analysis, I would like to display a graph which reflects the average industry stock returns and inflation to provide some visual scrutiny of the relationship between these variables. Although, my OLS result is based on the inference made from individual firms' stock returns response to independent variables, these graphs also demonstrate the negative relationship identified from my studies:





It is by no means clear that the stock returns have a strong noise while inflation is more stable. However, the negative relationship is can be grasped with a careful look especially after 2021. During the Covid-19 breakout stock returns were extremely sensitive and over reactionary to the crisis, especially the energy industry, which proves the impact of other observable and unobservable variables like investors' sentiment, inefficiency of market, inside trading and other array of intricate factors which are a subject for a further research.

Identification Issue

The identification strategy used in this research, while robust, faces several potential challenges in establishing the causal relationship between macroeconomic variables and real stock returns. One of the primary concerns is endogeneity, where independent variables such as inflation or expected inflation may be correlated with unobserved factors influencing stock returns. This issue can arise due to reverse causality, where stock market performance might influence macroeconomic variables like inflation expectations or money supply, or from omitted variables such as firm-specific characteristics or global economic shocks that simultaneously affect both stock returns and the macroeconomic indicators. To address endogeneity, future research could incorporate instrumental variable (IV) techniques, using external instruments like global commodity prices or lagged macroeconomic variables. Additionally, dynamic panel models such as the Generalized Method of Moments (GMM) could help mitigate these issues by leveraging internal instruments derived from lagged values of the variables.

Measurement error is another concern, particularly in variables like expected inflation, which often rely on proxies or estimates. These inaccuracies may attenuate the estimated coefficients and reduce their significance. Future projects could enhance the reliability of these measures by incorporating survey-based inflation expectations or inflation-indexed bonds (TIPS), which offer more precise and granular data. Another limitation is the absence of granular firm-level controls in the current model. Although the analysis includes industry-level fixed effects, firm-specific variables such as leverage, profitability, or market volatility are not accounted for, despite their significant role in influencing stock returns. Including these firm-level variables in future research would provide a richer and more detailed understanding of the dynamics at play.

The model also assumes immediate effects of macroeconomic variables on stock returns, potentially overlooking delayed responses. For example, inflationary pressures or changes in monetary policy may take time to influence stock performance. This temporal misalignment can be addressed by incorporating lagged values of independent variables or employing distributed lag models, which better capture the dynamics between macroeconomic indicators and stock returns. Moreover, while the fixed-effects approach effectively controls for unobserved heterogeneity across industries, it assumes that these effects remain constant over the study period. This assumption may not hold in the presence of structural changes, technological advancements, or regulatory shifts within industries. Future research could explore random-effects models or allow for time-varying fixed effects to accommodate evolving industry characteristics and enhance the generalizability of the findings.

Finally, the use of monthly data limits the ability to capture high-frequency variations or long-term trends in stock returns. Additionally, while the inclusion of industry-specific inflation rates is a novel aspect of this research, these rates may not fully reflect firm-level cost structures or input dependencies. Future studies could benefit from higher-frequency or longer-term data to improve granularity and robustness. Similarly, more precise inflation measures, such as firm-specific input cost indices, would enhance the accuracy of industry-level inflation effects.

Conclusion

This research provided critical insights into the relationship between inflation and stock performance, focusing on the firm and sectoral levels within the S&P 500. A key finding of the analysis is that real inflation has a statistically significant negative impact on stock returns, reinforcing the well-established *proxy hypothesis* proposed by Fama (1981). This hypothesis posits that inflation indirectly influences stock returns by reducing real economic activity, which subsequently lowers corporate profitability and stock valuations. The negative correlation observed in this study highlights how inflation erodes the real value of future cash flows, a core component of equity valuation models, and underscores the vulnerability of stock markets to inflationary pressures.

Expected inflation, while showing a positive relationship with stock returns, was not statistically significant in this study. This contrasts with prior research, such as Mandelker and Tandon (1985), which emphasized the predictive role of expected inflation in equity markets. One possible explanation for this discrepancy lies in the nature of the data used in this analysis. Unlike prior studies that focused on yearly expected inflation, this research incorporated monthly expected inflation. Monthly expectations may not exert the same influence on stock returns as yearly expectations, as short-term inflation forecasts might have less of a direct impact on long-term investment decisions or equity pricing.

Additionally, stock returns may be less sensitive to monthly inflation expectations because market participants could have already factored short-term inflationary trends into prices, effectively diminishing their observable impact. This aligns with the notion that markets are forward-looking and efficiently incorporate publicly available information into pricing. Another plausible reason could be the inherent difficulty in accurately estimating monthly expected inflation, which introduces noise into the analysis and reduces its statistical significance.

The divergence between this study's findings and earlier literature underscores the need for further exploration of time horizons in inflation expectations and their varying effects on stock

performance. It highlights the importance of considering the temporal dimension of inflation measures in financial research to better understand their predictive power in equity markets.

A novel aspect of this research was the inclusion of industry-specific inflation rates to capture their unique impact on sectoral stock returns. Interestingly, the results indicated that industry-specific inflation rates were statistically insignificant in influencing stock returns. This finding diverges from studies such as Eldomiaty et al. (2019), which identified sector-specific inflationary impacts. One plausible explanation is that industry-level inflation might be uniformly absorbed across firms within a sector or offset by other macroeconomic forces that dominate the effects of localized inflationary pressures. Alternatively, this result may stem from the difficulty in capturing accurate industry-specific inflation rates for certain sectors, particularly those with heterogeneous cost structures. Nonetheless, this approach remains a significant contribution, emphasizing the need for further exploration of localized inflationary dynamics, especially over varying time horizons.

The analysis confirmed the significant and positive relationship between money supply growth and stock returns, aligning with the findings of Eldomiaty et al. (2019) and reaffirming the role of liquidity in driving equity performance. Similarly, industrial production exhibited a robust positive relationship with stock returns, underscoring its importance as a proxy for real economic activity, consistent with Fama's (1981) framework. These results emphasize the pivotal role of real activity and liquidity in shaping stock market outcomes.

The positive and statistically significant relationship between Treasury bill rates and stock returns presented an interesting divergence from classical theories, which often predict a negative relationship. This result likely reflects the broader economic environment during the study period, where rising short-term rates were associated with economic expansion and improved investor confidence, thus boosting equity performance.

The sectoral analysis also revealed distinct sensitivities to macroeconomic variables. For instance, the IT sector demonstrated significantly higher returns relative to the Energy sector, reflecting its responsiveness to macroeconomic conditions such as liquidity and innovation cycles. In contrast, the Health Care sector's returns were not significantly different from the Energy sector, suggesting its relatively stable and defensive nature against economic fluctuations. These differences underscore the varying influences of macroeconomic forces across industries and the importance of tailoring investment strategies to specific sectoral contexts.

This study contributes to the literature by providing nuanced insights into the interplay between inflation, other macroeconomic variables, and stock performance at both the firm and sectoral levels. The findings offer actionable implications for investors aiming to hedge against inflation risks in diversified portfolios and for policymakers seeking to understand the broader economic implications of inflation on financial markets.

While the results align with several established theories, the weak significance of some variables, such as expected inflation and industry-specific inflation, raises questions about their practical impact on monthly stock returns. This outcome may reflect the difficulty in capturing the full scope of inflation expectations in shorter timeframes, as well as the potential influence of omitted variables such as financial risk ratings, market volatility, and firm-specific characteristics. These limitations highlight the potential for omitted variable bias, as other unobserved factors likely play a role in influencing individual stock returns.

Future research could address these gaps by incorporating additional variables such as global economic shocks, volatility indices, or firm-specific measures, and by leveraging dynamic modeling approaches to better capture forward-looking market behavior. Expanding the scope to include a larger number of sectors and companies would also enhance the robustness of the findings, offering a more comprehensive view of broader market dynamics. By addressing these challenges, future studies could provide deeper insights into the complex relationships between macroeconomic variables and equity markets, particularly in the context of inflation and sectoral differences. Ultimately, this research demonstrates that while inflation and macroeconomic variables significantly influence stock performance, their effects are multifaceted and vary across industries, providing a richer understanding of inflation's role in financial markets.

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Data Sources:

1. Monthly Stock Returns:

- Stock price and return data for the S&P 500 companies were sourced from Nasdaq, providing historical data from December 2014 to June 2024.
- Website: <https://www.nasdaq.com>

2. Inflation Metrics:

- **General Inflation and Industry-Specific Inflation Rates (Energy and Health Care):**
 - Sourced from the U.S. Bureau of Labor Statistics (BLS).
 - Website: <https://www.bls.gov>
- **IT Industry-Specific Inflation:**
 - Derived from the Federal Reserve Economic Data (FRED): "Consumer Price Index for All Urban Consumers: Information Technology, Hardware, and Services in U.S. City Average (CUSR0000SEEE)."
 - Website: <https://fred.stlouisfed.org>

3. Macroeconomic Variables:

- **Expected Inflation:** Extracted from the Federal Reserve Economic Data (FRED).
 - Website: <https://fred.stlouisfed.org>
- **3-Month Treasury Bill Rates (TB-3):** Provided by the Federal Reserve Economic Data (FRED).
 - Website: <https://fred.stlouisfed.org>
- **Money Supply (M2):** Broad money supply data from FRED.
 - Website: <https://fred.stlouisfed.org>
- **Industrial Production and Gross National Product (GNP):** Both variables were also obtained from FRED.
 - Website: <https://fred.stlouisfed.org>